Atalaya: A Calibrated Multi-Evidence Affinity Graph over an Open-Data Catalog, with an Empirical False-Discovery Gate

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***Abstract*—Open-data portals publish thousands of datasets as a flat searchable list, but the relational knowledge that makes the data useful, which datasets can be joined, which describe the same thing, and which actually correlate when aligned by a common key, stays implicit and is infeasible to recover by hand. Existing dataset-discovery tools each rank relatedness on a *single* signal (value containment for joinability, embedding cosine for semantic search, statistical association for correlation mining), and each signal is individually misleading: two datasets can be embedding-similar yet share no join key, joinable on *year* yet describe unrelated phenomena, or correlate spuriously through a shared driver. We present Atalaya, a system that harvests a national open-data catalog (1,017 datasets from Chile's Data Observatory; a 303-dataset, 24.8 GB government subset mirrored and profiled in full), mines five orthogonal kinds of cross-dataset relation into an explorable knowledge graph, and ranks pairwise relatedness with a *calibrated multi-evidence affinity* that fuses the evidence with standard empirical-CDF calibration and reliability weighting. The affinity fuses three orthogonal evidences into one interpretable $[0, 1]$ score after passing each through its *empirical null distribution*, so the score means "stronger than chance" rather than "large raw number", and it down-weights any evidence that contradicts the others. Two design choices make the graph trustworthy rather than merely plausible. First, the correlation edges pass an *empirical false-discovery gate*: on the same alignments with the keys shuffled, 343 candidate correlations yield 0 survivors under a permutation null with Benjamini–Hochberg control (empirical false-discovery rate 0.00), so the 24 correlations reported on real data are believable. Second, the semantic encoder is measured honestly against a classical lexical baseline, and we report the modest true margin ($+0.014$ top-5 neighbour-theme coherence over TF–IDF) rather than overselling it. Every edge carries its evidence in the graph database, the whole affinity is pure `numpy` so the browser recomputes it live as an analyst reweights the evidences, and all figures are regenerated deterministically from the committed artifacts. We give the method, the adversarial validation, and an explicit account of what the graph does and does not license.**

***Index Terms*—Dataset discovery, data integration, joinability, knowledge graph, semantic search, correlation mining, false-discovery rate, null calibration, open data, reproducibility.**

I. Introduction

A NATIONAL open-data portal is a public good that undersells itself. Chile's Data Observatory, like most portals built on the FAIR principles [1], exposes over a thousand datasets as a flat, searchable list with per-dataset metadata. The list answers *what* datasets exist. It does not answer the question an analyst actually has, which is *how they connect*: which two tables can be joined and on what column, which describe the same phenomenon under different names, and which carry indicators that genuinely move together when aligned by *comuna* (municipality) or region. That relational structure is the difference between a thousand isolated files and a usable data fabric, and at the scale of a national catalog it cannot be recovered by hand: the number of candidate dataset pairs is quadratic in the catalog size, and most pairs are unrelated.

The database and information-retrieval communities have built strong tools for pieces of this problem. Data discovery and augmentation systems such as Aurum [2] and Auctus [3] index a corpus and, given a query table, return joinable or unionable tables; joinability at scale rests on estimating value-set containment from compact sketches (MinHash [4], sharpened by Lazo [5] and indexed by LSH Ensemble [6]). Semantic catalog search embeds dataset text with a sentence encoder [7], [8] and ranks by cosine similarity; Google Dataset Search [9] operationalises this at web scale. Correlation miners look for statistical association between aligned indicator columns. Each of these is excellent at its one signal, and each is, on its own, an unreliable notion of "related". A pair can be embedding-similar because both datasets discuss "population" in prose, yet share no joinable key. A pair can be perfectly joinable on *year* and describe entirely unrelated phenomena. A pair can correlate strongly and spuriously because both track a common driver such as population or region [14]. Worse, a large catalog is a multiple-testing minefield: mine thousands of candidate correlations without correction and a handful will look significant by chance [13].

Atalaya's premise is that a trustworthy relatedness graph must do three things a single-signal ranker does not. It must **fuse orthogonal evidences** so the strengths of one cover the blind spots of another. It must **calibrate each evidence against a null model** so a score reflects surprise relative to random pairs, not the raw magnitude of a signal whose scale is arbitrary. And it must be **honest and auditable**: every edge should carry the evidence that justifies it, the statistical edges

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Preprint, 2026. Source, committed artifacts, and a live instance are open (MIT): https://github.com/fsantibanezleal/CAOS_ATALAYA; https://atalaya.fasl-work.com.

should survive an adversarial negative control, and a learned component should be measured against a classical baseline with the real margin reported. This paper contributes:

1) A **calibrated multi-evidence dataset-affinity score** (Section V) that maps three orthogonal raw evidences through their empirical null distributions and fuses them with reliability weights that discount contradicting evidence. The score is in $[0, 1]$, monotone, interpretable term-by-term, and cheap enough (pure `numpy`) to recompute live in the browser under user reweighting.
2) An **honest relation-mining pipeline** (Section IV) that produces five orthogonal edge kinds over a real national catalog, each recording its own evidence, with a correlation miner built from a seeded permutation null and Benjamini–Hochberg false-discovery control (a first-order partial-correlation confounder check is implemented but is not applied to the committed run).
3) An **adversarial validation** (Section VI) centred on an *empirical false-discovery gate*: re-mining correlations on shuffled key alignments yields an empirical false-discovery rate of 0.00 (0 of 343 candidate correlations survive), and the semantic encoder is benchmarked against a TF–IDF lexical baseline with the modest true gain reported.
4) A **reproducible, explorable instance** over 1,017 datasets (Section VII), with all derived artifacts committed and every figure in this paper regenerated deterministically from them.

We are deliberate about scope (Section VIII): the graph reports where datasets *can* be related and how strongly relative to chance; it does not assert causation, and its tunables are set for one government-Chile catalog. The value is not a claim of discovery but a method for turning a flat catalog into an auditable relational map without inviting the reader to believe more than the evidence supports.

II. Related Work

Dataset discovery and augmentation. Aurum [2] builds an enterprise knowledge graph of columns linked by content similarity and primary-key/foreign-key relationships; Auctus [3] is a dataset search engine oriented to data augmentation, ranking candidate tables that join or union with a user table. Both target the "given this table, find related tables" query. Atalaya differs in producing a *catalog-wide* undirected relatedness graph with a single fused score per pair, rather than a per-query ranked list on one criterion, and in calibrating that score against a null.

Joinability from sketches. Estimating the containment of one column's value set in another underlies joinability. MinHash [4] gives an unbiased Jaccard estimate from fixed-length signatures; Lazo [5] jointly estimates Jaccard and containment with a cardinality-aware correction; LSH Ensemble [6] indexes containment search at internet scale. Atalaya uses MinHash-signature containment on declared entity keys as its joinability evidence and treats it as the most reliable of the three fused signals (a shared key is a fact, not a guess).

Semantic catalog search. Sentence encoders [7], and in particular distilled compact models such as MiniLM [8], embed dataset titles and descriptions so that cosine similarity captures topical relatedness; Google Dataset Search [9] shows the approach at web scale. We use a MiniLM embedding as the semantic evidence, but treat raw cosine as *one* calibrated input rather than the ranking itself, and we measure it against a classical lexical baseline [10] rather than assuming it wins.

Data profiling and relationship mining. Profiling relational data [11] (types, keys, cardinalities, dependencies) is the substrate for any relationship mining; our harvesting stage profiles every downloadable table before any edge is drawn. For statistical relationships, Spearman rank correlation [12] is robust to the monotone nonlinearities and outliers typical of government indicators; controlling the false-discovery rate across a family of tests [13] is essential at catalog scale; and partial correlation [14] tests whether an apparent association survives removing a common driver. Atalaya combines all three, and adds an empirical negative control that most correlation miners omit.

Projection. The map view projects the multilingual embedding space with principal component analysis [15]; the graph itself is stored and queried with a high-performance library [16].

III. The Catalog and Its Harvesting

Atalaya targets Chile's Data Observatory catalog. The harvesting stage enumerates every dataset, records its metadata (title, description, theme, publishing organisation, license, declared entity keys), and, for the government direct-file tier, downloads and profiles the underlying tables in full. As of the committed snapshot (measured 2026-07-02), the catalog holds **1,017** datasets; the Tier-A government subset mirrored and profiled in full comprises **303** datasets, **1,382** files, and **24.8 GB** (26,604,576,347 bytes, recorded in `harvest_report.json`). Profiling each table yields, per column, its type, null fraction, cardinality, a sample of values, and, for candidate key columns, a MinHash signature for containment estimation. Two representations are computed per dataset and cached as compact artifacts: (i) a *semantic text* (title, description, keywords, column names) and its MiniLM embedding [8], and (ii) the set of *declared entity keys* it is aligned on, drawn from a small government-Chile entity vocabulary (municipality code *comuna_cut*, region, year, latitude/longitude). The entity keys strong enough to anchor a join or a correlation are *comuna_cut* and region; these are the alignment keys used throughout.

All later stages consume only these committed artifacts; no stage in this paper requires the network or a live recompute. This is deliberate: the derived graph, the evaluation metrics, and the figures are all reproducible from the repository at a fixed commit.

IV. Five Orthogonal Relation Evidences

The relation-mining stage draws five kinds of undirected edge, each with explicit evidence recorded on the edge so that the graph is auditable rather than opaque. Ordered from cheapest prior to strongest structural evidence:

- **SAME_SOURCE**: the two datasets share a publishing organisation. A cheap prior, given a weak weight; it captures administrative provenance, not content relatedness.
- **SEMANTICALLY_SIMILAR**: cosine similarity of the two MiniLM embeddings exceeds a threshold (≥ 0.45), kept as the top-k ($k = 8$) neighbours per dataset. Topical relatedness of the described content.
- **SPATIALLY_OVERLAPS**: both datasets are municipality/region-keyed or carry coordinates in the same area. Geographic co-support, a precondition for a meaningful spatial join.
- **JOINABLE_ON**: the two datasets share a declared entity key whose value sets have high MinHash containment (≥ 0.5), i.e. a candidate foreign-key link with the exact columns to join on recorded. The strongest structural evidence: a shared key is a fact.
- **CORRELATES**: two indicator columns, aggregated to a shared key, correlate by Spearman's ρ [12]; the correlation survives a seeded permutation null and Benjamini–Hochberg control [13] across the whole candidate family. The most expensive edge to mine, and the one whose interpretation demands the most care (see Section VIII).

The statistics for the last kind are implemented in pure `numpy`/`math` (rank transform, Pearson on ranks for Spearman, an add-one-smoothed two-sided permutation p-value with a seeded shuffle, the Benjamini–Hochberg step-up, and first-order partial rank-correlation). This keeps the miner light enough to run identically offline and in the browser, and avoids a t-approximation that would assume bivariate normality government indicators rarely satisfy. The candidate family is bounded (at most 6 indicators per dataset, a global cap on the number of tests, both logged if hit) so the multiple-testing correction is over a fixed, honest denominator. Every accepted edge additionally receives the fused affinity of Section V as a dataset-level summary, so the graph carries both the specific evidence and the calibrated overall strength.

V. Calibrated Multi-Evidence Affinity

The core contribution is a single relatedness score that fuses the three orthogonal evidences of Section IV (we fuse semantic, joinability, and statistical; the two remaining edge kinds are provenance and geographic preconditions, not relatedness signals). Let a dataset pair (A, B) have raw semantic cosine s, raw join containment j, and raw statistical strength t. Two problems must be solved before these can be added: their scales are not comparable (a cosine of 0.6 and a containment of 0.6 do not mean the same thing), and a large raw value is not the same as a *surprising* one.

Null calibration. We solve both by mapping each raw signal through its *empirical null cumulative distribution*. For evidence $e \in \{s, j, t\}$ we draw a background sample of the raw signal over random dataset pairs and fit its empirical CDF $\widehat{F}_e$; the calibrated evidence is its percentile,

$$f_e \;=\; \widehat{F}_e(e) \;=\; \Pr\left[\, E_e \leq e \,\middle|\, \text{random pair} \,\right] \;\in\; [0, 1], \quad (1)$$

computed by a binary search in the sorted background sample. After calibration, f_e is directly interpretable as "this pair's evidence e is stronger than a fraction f_e of random pairs", on a common $[0, 1]$ scale for all three evidences. The background samples (small sorted arrays) are fit offline and shipped with the graph so the calibration is reproducible and available in the browser.

Reliability weighting. Orthogonal evidences should not be trusted equally when they disagree. A high semantic similarity with no joinability is often topical coincidence (two datasets both mention "education"), whereas a strong shared key with no semantic echo is still a real structural link. We encode this with gentle reliability multipliers in $[0.5, 1]$ that raise an evidence when the others agree and lower it when they do not:

$$r_s = 0.6 + 0.4\,\max(f_j, f_t), \quad (2)$$

$$r_j = 0.85 + 0.15\,\max(f_s, f_t), \quad (3)$$

$$r_t = 0.5 + 0.5\,f_j. \quad (4)$$

Joinability starts high (a shared key is a fact, lightly boosted by agreement); statistical association is trusted only in proportion to a joinable alignment (without a real alignment the correlation is dubious); the semantic term is fully trusted only with some structural or statistical support.

Fusion. With base evidence weights w_s, w_j, w_t (default $0.34, 0.40, 0.26$; user-adjustable), the fused affinity is the reliability-weighted, null-calibrated convex combination

$$S(A, B) \;=\; \frac{w_s r_s f_s + w_j r_j f_j + w_t r_t f_t}{w_s r_s + w_j r_j + w_t r_t} \;\in\; [0, 1], \quad (5)$$

with $S = 0$ when the denominator vanishes. The result is monotone in each calibrated evidence, invariant to the overall scale of the weights (the denominator renormalises), and, crucially, *reported term by term*: every affinity edge stores f_s, f_j, f_t and the effective weights $w_e r_e$, so a reader can see *why* two datasets are linked and never confronts an opaque number. Figure 1 shows the pipeline on a worked pair. Because the whole computation is a handful of `numpy` operations over small per-pair vectors and shipped background arrays, the browser recomputes S for the visible subgraph in real time as the analyst moves the evidence weights, which turns the score from a fixed ranking into an interactive instrument for asking "rank these datasets as if I cared mostly about joinability" versus "…mostly about statistical echo".

VI. Adversarial Validation

The honest question about a relatedness graph is not "did we find links" but "are the links stronger than a null world". We answer it with four adversarial or leakage-safe checks, all computed in the evaluation stage and committed to `metrics.json`.

The empirical false-discovery gate (the key check). The correlation edges are the ones most easily faked by multiple testing, so we subject them to a negative control. We take the exact indicator series and key alignments used to mine correlations, *shuffle* the key labels of each series to destroy any true alignment, and re-run the identical pipeline: Spearman, seeded permutation null, Benjamini–Hochberg control at $q = 0.05$. A trustworthy miner should find essentially nothing on shuffled data. It finds nothing: of 343 candidate pairs that clear

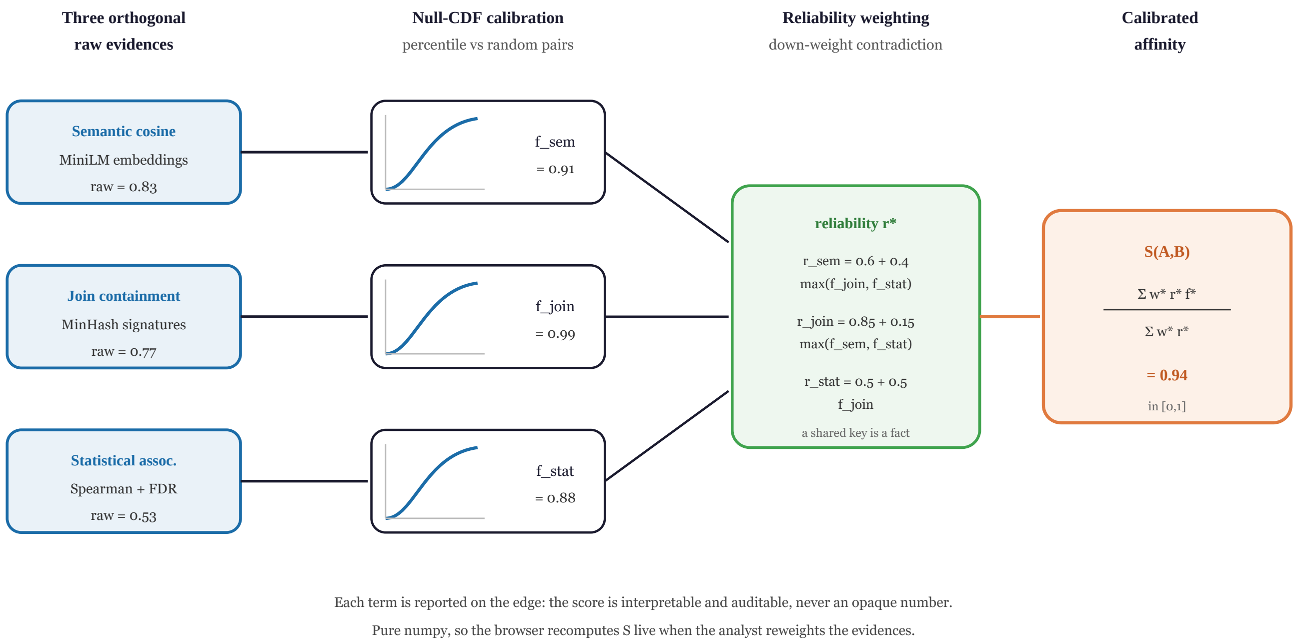

Fig. 1. The calibrated multi-evidence affinity. Three orthogonal raw evidences (semantic embedding cosine, MinHash join containment, statistical association) are each mapped through their *empirical null CDF*, i.e. the percentile of the raw signal against a background sample of random dataset pairs, so a value near 1 means "far stronger than a random pair" regardless of the raw scale. Reliability weights then discount an evidence that contradicts the others (a high semantic match with no joinability is a weak reason to link; a shared key is intrinsically reliable), and the fused score is the reliability-weighted, null-calibrated mean, reported term-by-term. The whole computation is pure `numpy`, so an analyst can reweight the evidences and see the graph re-rank live in the browser.

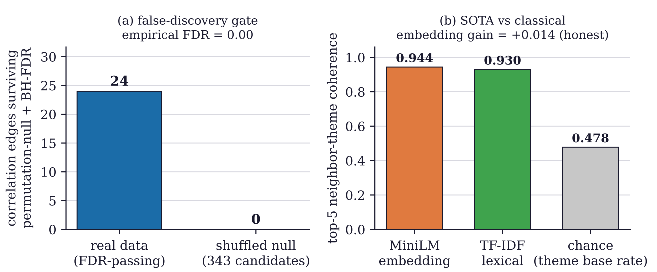

Fig. 2. Adversarial validation, from the committed metrics. **(a)** The empirical false-discovery gate: the correlation miner accepts 24 edges on the real data, but re-running the *identical* test family on *shuffled* key alignments (breaking any true association) leaves 0 survivors out of 343 candidates, an empirical false-discovery rate of 0.00. **(b)** Honest SOTA-versus-classical: the MiniLM embedding reaches 0.944 top-5 neighbour-theme coherence against 0.930 for a TF–IDF lexical baseline over the same text, both far above the theme base rate of 0.478; the true embedding gain is a modest +0.014, reported rather than overstated.

the effect-size screen on the shuffled data, 0 survive the null and FDR control, an empirical false-discovery rate of 0.00 (Fig. 2a). This is the empirical warrant for believing the 24 correlations reported on real data; without it, "survived FDR" is a promise, not a measurement.

Honest SOTA-versus-classical. It is tempting to justify a learned embedding by asserting it beats a lexical baseline; we measure the gap instead. Over the identical set of datasets we compute top-5 neighbour-theme coherence (the fraction of a dataset's five nearest neighbours sharing its theme, a leakage-free proxy for embedding quality) twice: once from MiniLM cosine, once from a TF–IDF lexical similarity [10] over the same text. The embedding reaches 0.944 and the lexical baseline 0.930, both far above the theme base rate of 0.478 (Fig. 2b). The SOTA gain is a real but modest +0.014; reporting it honestly is the point, and it justifies using the embedding as *one* calibrated evidence rather than as the whole ranking.

Joinability sanity and semantic coherence. Two further checks bound the graph's internal consistency. Every **JOINABLE_ON** edge should connect datasets that share a declared entity key; the measured fraction is 1.000 over all 117 join edges (a drop would signal a bug). And the semantic neighbourhood is coherent: across 4,753 scored top-k neighbour slots, 0.940 share the source dataset's theme, far above the base rate. Together with an isolated-node fraction of 0.0049 (almost every dataset participates in at least one relation), these establish that the graph is connected, internally consistent, and not an artifact of a leaky test.

VII. Results: The Relatedness Graph

The committed graph over the 1,017-dataset catalog has 14,413 edges (Fig. 3): 6,002 SAME_SOURCE, 3,948 SEMANTICALLY_SIMILAR, 269 SPATIALLY_OVERLAPS, 117 JOINABLE_ON, 24 CORRELATES, and 4,053 AFFINITY summary edges. The composition itself is part of the honesty: the vast majority of raw edges are cheap priors and topical similarities, while the structurally hard evidence (joins and FDR-passing correlations) is a small, carefully gated minority. The affinity score exists precisely so that this mixture can be collapsed into one calibrated number per pair without losing the distinction, since a pair with a strong join and a

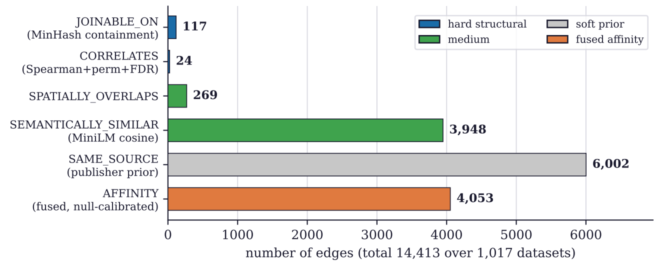

Fig. 3. Edge-kind composition of the committed graph (14,413 edges over 1,017 datasets), coloured by evidence strength. The graph is honest about its own evidence hierarchy: a small number of *hard* structural links (117 joins, 24 FDR-passing correlations), more *medium* semantic and spatial evidence, a large but *soft* same-source prior, and the 4,053 fused, null-calibrated affinity summaries. The interface renders the same hierarchy so an analyst is never invited to read a cheap prior as a strong relation.

statistical echo scores far above a pair that is merely same-source.

Three concrete edges illustrate what the graph surfaces. The highest-affinity pair, *Nombramientos Sistema de Alta Dirección Pública* and *Convocatorias Sistema de Alta Dirección Pública* (appointments and calls of the senior public-management system), scores $S = 0.999$ with $f_s = 0.999$, $f_j = 1.000$, $f_t = 0.999$: two tables of the same administrative process, near-identical in text, perfectly joinable, and statistically aligned, exactly the pair a human would call "obviously the same family". A representative correlation edge links *Establecimientos educacionales* and *Establecimientos de salud vigentes* (educational and active health establishments) at $\rho = 0.528$, permutation $p = 0.001$ (accepted under BH-FDR at $q = 0.05$), over $n = 324$ municipalities aligned on *comuna_cut*. The aligned columns here are administrative region and health-authority codes, so this ρ reflects a shared north-to-south regional ordering rather than establishment co-location; we report it as a code-level association, not a substantive relation, and not a cause. A representative join edge connects two municipality-keyed tables with MinHash containment 0.769 on their *comuna* columns, recording the exact columns and set sizes so the join is executable, not merely suggested. Each of these is inspectable in the live instance with its full evidence.

VIII. Honest Scope and Limitations

The graph reports *potential* relatedness with calibrated, auditable strength; it does not assert causation and it is tuned for one catalog. Five limitations are worth stating plainly.

Association, not causation. A CORRELATES edge means two columns move together across municipalities after a permutation null and FDR control; the committed correlation run is *not* partial-correlation screened (a first-order confounder check is implemented but not applied). It does not identify a mechanism, and a confounder can still induce it. The interface labels these edges as associations.

Administrative-code correlations. The correlation miner selects any numeric-castable column, so a share of the 24 CORRELATES edges align administrative or geographic *code* columns (region, service and entity codes) rather than substantive indicators, some over as few as eight aligned units, and a few are near-tautological code-versus-code alignments. These should be read as code-level alignments, not indicator relationships; screening out identifier and code columns before correlating, and applying the partial-correlation check, is the clear next fix.

Evidence hierarchy is real and unequal. Of the 14,413 edges, only $117 + 24$ are structurally hard; the rest are priors or topical similarities. The affinity weighting encodes this, but a user who reweights all mass onto the semantic evidence will get a graph dominated by topical coincidence. The reliability multipliers mitigate but do not eliminate this, and the interface keeps the raw evidence visible for exactly this reason.

Catalog-specific tuning. The alignment keys (*comuna_cut*, region), the government-Chile entity vocabulary, and the thresholds (semantic ≥ 0.45, containment ≥ 0.5, $|\rho| \geq 0.35$, minimum overlap 8, $q = 0.05$) are set for this catalog. The *method* transfers, but the exact tunables would need re-fitting for another portal, and the null backgrounds would need re-sampling.

Deliberately light statistics. The correlation miner uses Spearman with a permutation null rather than a heavier dependence measure, and the affinity uses first-order (not full) reliability interactions. This is a conscious trade for a stack that runs identically offline and in the browser; a server-side variant could use richer dependence estimators at the cost of the live lane.

Coverage. Full table profiling covers the 303-dataset government direct-file tier; datasets outside it are catalogued and embedded from metadata but do not contribute join or correlation evidence, so their edges rest on the semantic and same-source signals only. The graph is honest about this: such datasets simply carry no hard structural edges.

IX. Reproducibility and Availability

Every number in this paper comes from artifacts committed to the repository: the graph counts (`graph.json.counts`), the evaluation metrics (`metrics.json`), the harvest report (`harvest_report.json`), and the per-case top-evidence artifacts. The committed `graph.json` is a top-4,000-by-weight web payload carrying edge endpoints and weights only; the full per-edge evidence lives in the out-of-git graph database. The three figures are regenerated deterministically by a single script from those artifacts (the two data figures) and from a hand-authored, theme-aware vector schematic (the method figure), with no network access. The pipeline is staged, seeded, and gated by determinism and data-contract tests. The relatedness statistics and the affinity are pure `numpy`/`math`, so the same code runs in the offline pipeline and in the browser live lane; the graph is also queryable from an agent via an included MCP server. Source, artifacts, and a live instance are open under the MIT license at https://github.com/fsantibanezleal/CAOS_ATALAYA and https://atalaya.fasl-work.com.

X. Conclusion

Atalaya turns a flat national open-data catalog into an auditable relatedness graph by fusing three orthogonal evidences into a single calibrated score, and by holding that graph to an adversarial standard rather than a plausibility standard. The calibration against empirical null distributions makes a score mean "stronger than chance"; the reliability weighting makes contradicting evidence count for less; the term-by-term reporting makes every edge auditable; and the empirical false-discovery gate (0 of 343 shuffled correlations survive) makes the statistical edges believable in a way an uncontrolled miner cannot claim. The honest SOTA-versus-classical margin ($+0.014$) is reported rather than oversold, in keeping with the system's premise that a data map is only useful if it can be trusted. The method is not tied to this catalog; what transfers is the recipe of fuse, calibrate against a null, weight by reliability, and gate against a shuffled negative control, for anyone building a relatedness graph over a large, heterogeneous collection of datasets.

Appendix A
Configuration and Null Construction

For exact reproduction, the relation-mining tunables are: semantic top-$k = 8$ with minimum cosine 0.45; join minimum MinHash containment 0.50 on the anchor keys *comuna_cut* and region; correlation minimum shared key values 8, minimum $|\rho|$ 0.35, at most 6 indicator columns per dataset, a global cap of $40{,}000$ correlation tests (logged if reached), and Benjamini–Hochberg control at $q = 0.05$. The permutation null uses a seeded generator ($1{,}000$ permutations in the miner, 500 in the negative control) with add-one smoothing so a reported p-value is never exactly zero. The empirical null CDF for each affinity evidence is fit offline by sampling the raw signal over random dataset pairs, sorting the sample, and evaluating the percentile by binary search at query time; the sorted sample is shipped with the graph so the calibration is identical offline and in the browser. Default affinity weights are $w_s = 0.34$, $w_j = 0.40$, $w_t = 0.26$ and are renormalised by the fusion denominator, so only their ratios matter.

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